QUESTION 1

A CPU has 256KB of SRAM connected to it starting at address 0x00100000. What is the address of the last byte in the SRAM?

Options

A. 0x00110000

B. 0x00103FFF

C. 0x000FFFFF

D. 0x0013FFFF

Clear Selection

QUESTION 2

Control registers for a UART peripheral start at address 0x41001000. There are 8 control registers, each 16 bits wide. What is the address of the last register?

Options

A. 0x41001016

B. 0x4100101C

C. 0x41001020

D. 0x41001014

Clear Selection

QUESTION 3

Common Statement for Questions: 3 to 4

For the following Questions, Read from pg. 145 of RM0490 Reference manual.
Reference manual Link: Link

What will be the result of performing a READ operation from a GPIOX_ODR register?

Options

A. The current value stored in this register, which reflects the output state of the GPIO pins associated with GPIOX.

B. An error, as GPIOX_ODR registers are write-only

C. The current value stored in this register, which reflects the input state of the GPIO pins associated with GPIOX.

D. None of the above

Clear Selection

QUESTION 4

Common Statement for Questions: 3 to 4

For the following Questions, Read from pg. 145 of RM0490 Reference manual.
Reference manual Link: Link

What will be the result of performing a WRITE operation to a GPIOx_IDR register?

Options

A. The corresponding GPIO pins will be set to high (logic level 1)

B. The data written will be stored temporarily and then overwritten by subsequent reads

C. The operation will be ignored because it is input data register and used only for reading port values

D. The microcontroller will enter a low-power mode

Clear Selection

QUESTION 5

Common Statement for Questions: 5 to 6

CPU has supply voltage of 3.3V. You use a pullup resistor of 1000 ohms on a GPIO port.
How much power is dissipated in the resistor when the input is not connected?

Options

A. 0.033 mW

B. 0 W

C. 3.3 mW

D. 3.3 W

Clear Selection

QUESTION 6

Common Statement for Questions: 5 to 6

CPU has supply voltage of 3.3V. You use a pullup resistor of 1000 ohms on a GPIO port.
How much current flows through the resistor when the input is connected to ground?

Options

A. 3.3 mA

B. 0 mA

C. 3.3 µA

D. 3.3 A

Clear Selection

QUESTION 7

You are using a CPU with a 3.3V supply voltage to drive a red LED with a forward voltage V_F of 2.1V. What should the value of the protection resistor be to ensure the current through the LED is not more than 20mA?

Options

A. 60 ohms

B. 56 ohms

C. 68 ohms

D. 100 ohms

Clear Selection

QUESTION 8

What is the purpose of HAL (Hardware Abstraction Layer) in embedded systems?

Options

A. Abstract away the hardware details such as pin numbers or how to set/reset pins

B. Simulating the system without actual hardware

C. Write programs in languages other than C

D. Use GPIOs for sampling analog signals

Clear Selection

QUESTION 9

Which of the following statements about UART communication are true? Select all that apply.

Options — Select all that apply

- [ ] A. UART can only be used for communication between two CPUs.
- [ ] B. UART operates exclusively with peripherals and cannot communicate between CPUs.
- [ ] C. UART is a widely used communication protocol for serial communication.
- [ ] D. UART allows bidirectional data transfer between devices.

Clear Selection

QUESTION 10

Using 8N1 protocol at 9600bps, how much time will it take to transfer 100 bytes?

Options

A. 83.3 milliseconds

B. 104.2 milliseconds

C. 125 milliseconds

D. 156.25 milliseconds

Clear Selection